CS6P008 - Deep Learning Lab

Lab 9, Date: April 13, 2026

Topic(s): Regression Using CNN

Timing: 10:00 AM to 01:00 PM | Max Points: 10

Instructions

1. Provide commented, indented code. Variables should have meaningful names.
2. Submit two .ipynb files.
   The names should be `[student name]_lab[number]_1.ipynb` (before completion of the lab at 1 PM) and `[student name]_lab[number]_2.ipynb` (your final code file by the deadline).
3. Read the questions carefully before answering. If a question asks to follow a particular approach or to use a specific data structure, then it must be followed.
4. Write questions in separate text blocks in Jupyter/Colab Notebook before the code blocks containing answers.
5. All plots should have appropriate axis labels, titles, and legends.

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1. Given a dataset of colored images (CIFAR-10), the goal is to train a model that learns to colorize grayscale images. This will be achieved by:
   - Converting RGB images to grayscale for input (use Python code).
   - Using a pre-trained CNN (e.g., DenseNet-161) as an encoder.
   - Designing a custom decoder to generate RGB output.
   - Training the model with MSE loss and evaluating the output quality.

   ## Model Architecture (Refer to Figure 1)

   - **Encoder:** Pre-trained DenseNet-161, modified to accept 1-channel grayscale input.
   - **Decoder:** Custom CNN with transpose convolutions (also known as Deconvolution) to upsample and generate 3-channel RGB output.
   - **Loss Function:** Mean Squared Error (MSE) is used to measure the difference between the predicted RGB image and the ground truth.
   - **Optimizer**: Adam
   - **Learning Rate**: 0.001
   - **Epochs**: Train till convergence and use exponential learning rate decay with decay rate of 0.001

2. Answer the following questions based on your implementation and experimental observations:

- Plot training and validation loss vs epochs
- Explain how the model is able to colorize grayscale inputs with reasonable accuracy.
- For a random sample, plot the grayscale input, colorized output, and ground truth
- Log these metrics: MAE, MSE, RMSE, log(RMSE)

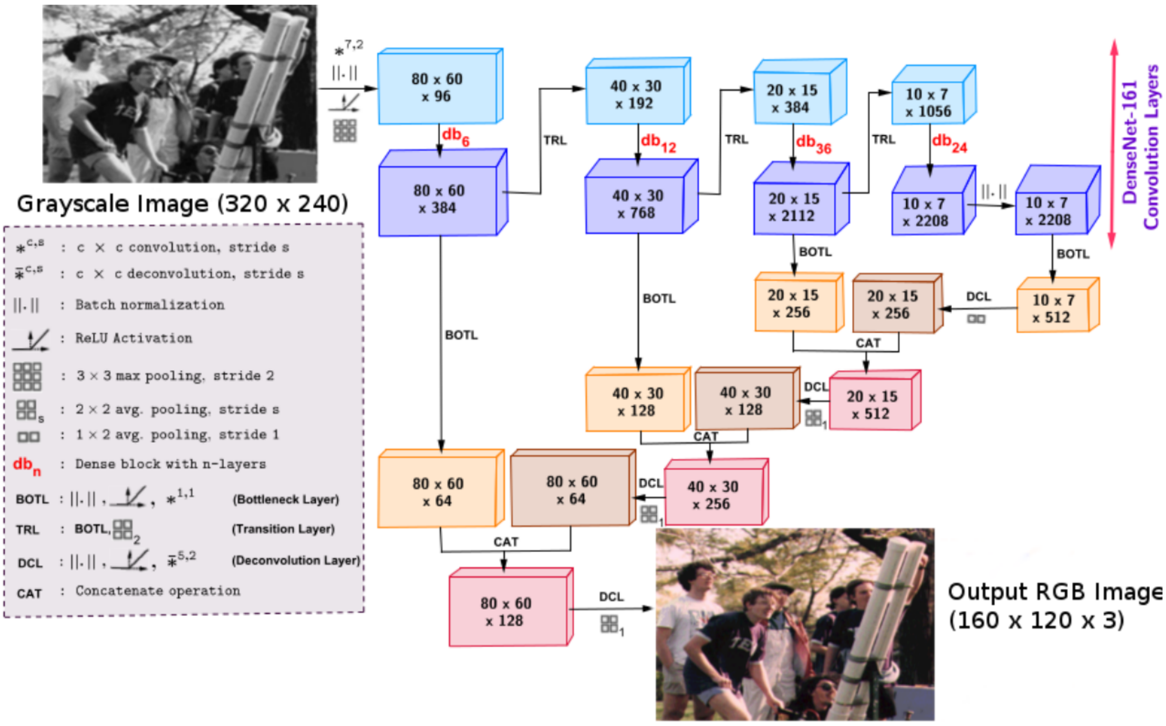

Figure 1: CNN Architecture Diagram